RTD Serial number: _____

CHILD REFERENCE TEST DEVICE – BUOYANCY TRACKING AND VERIFICATION

To achieve repeatability in human subject and manikin testing, the overall buoyancy and

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APPENDIX

RTD Serial number: _____

CHILD REFERENCE TEST DEVICE – BUOYANCY TRACKING AND VERIFICATION

To achieve repeatability in human subject and manikin testing, the overall buoyancy and distribution of buoyancy between the front a nd back of the RTD should be maintained within a tight tolerance as specified in Table 1.

Table 1 – SOLAS child RTD buoyancy and tolerance

Limit / Units Front buoyancy (1) (2) Back buoyancy (1) Total buoyancy (1) Buoyancy distribution (3)
Design / N 63 25 88 71.5% in front
Maximum / N 65.4 26.2 91.6 73% in front
Minimum / N 60.6 23.8 84.4 70% in front
(1) Values at or corrected to standard temperature and pressure.
(2) The left-to-right buoyancy distribution in the front inserts is to be within 1.3 N of each other.
(3) Buoyancy distribution is calculated by dividing the front buoyancy by the total buoyancy.

The buoyancy of a new RTD may exceed the al lowable tolerance range until the normal shrinkage or compression of the foam inserts stabilizes. Until the buoyancies of the foam inserts have stabilized, buoyancy and distribut ion should be checked at regular intervals

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shrinkage or compression of the foam inserts stabilizes. Until the buoyancies of the foam inserts have stabilized, buoyancy and distribut ion should be checked at regular intervals (perhaps weekly), and then at least monthly thereafter or whenever used for testing, whichever is longer (frequent use may require more frequent checks). Only RTD’s with buoyancies within tolerance should be used fo r certification testing. A data sheet to document the buoyancy and buoyancy distribution of the RTD is attached.

To check buoyancy tolerances, foam inserts n eed to be removed from the device. Take care that all trapped air is removed when checking buoyancy and that layers are maintained in their proper sequence when reinstalled (considerable effort would be needed to remove entrapped air if testing the intact device).

Adjustment of buoyancy: At the time of manufacture th e left-to-right distribution of buoyancy in the front inserts was adjusted to be within 1.3 N of each other. To achieve this tolerance, the layers were individually selected to achieve the cumulative insert buoyancy. If buoyancy of a new device exceeds the upper limits, one layer per

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this tolerance, the layers were individually selected to achieve the cumulative insert buoyancy. If buoyancy of a new device exceeds the upper limits, one layer per compartment may be altered or replaced to bring the unit into compliance. The test house may need to add make-up layers (see figure 3) from time to time to maintain the front-to-back and side -to-side insert tolerances. If the front buoyancy is under the minimum value, measure the buoyancy of the right and left sides so that the proper distribution of buoyancy (no more than a 1.3 N difference) between the right and left front panels can be maintained.
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Table 2 – SOLAS child RTD insert design buoyancies

Combined left front
insert (13 layers)
Combined right front
insert (13 layers)
Combined back
insert (11 layers)
Design (N) 31.5 31.5 25
S/N ____
Date:

RTD BUOYANCY DATA SHEET

RTD Serial number/identification: _____

Date Left front
buoyancy
(N)
Right
front

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SOURCE: MSC.200(80).pdf
insert (13 layers)

Combined back
insert (11 layers)
Design (N) 31.5 31.5 25
S/N ____
Date:

RTD BUOYANCY DATA SHEET

RTD Serial number/identification: _____

Date	Left front buoyancy (N)	Right front buoyancy (N)	Total front buoyancy (N)	Total back buoyancy (N)	Total buoyancy (N)	Buoyancy distribution (% in front)	Remarks
	(1)		(2)				

(1) Left and right front buoyancy need not be checked if distribution is within tolerance.
(2) If the temperature and pressure at the time of measurement is not at standard conditions, these

values should be corrected to standard temperature and pressure.

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Key
1 Trim upper right corner only fo r left insert layers A and B.

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